LINGVESSA
TECHNICAL SPECIFICATION
version 1.7

Owner-locked technical baseline; P0 authorized on 10.09.2026.

Field	Value
Identifier	LINGVESSA-TECH-SPEC
Version / date	1.7 / 09.09.2026
Status	OWNER-LOCKED IMPLEMENTATION BASELINE; owner lock 10.09.2026; P0 authorized.
Governing baseline	Formal Specification v1.7; Technical Specification v1.7; Technical Design v1.7; Curriculum Blueprint A1 v1.6; Reference Module Design v1.6; AI Integration Specification v1.4; Implementation Plan v1.4 (execution order only).
OpenAPI	OpenAPI Specification 3.1.2
Execution authority	Lingvessa_Implementation_Plan v1.4; P0 must be closed before P1.

Part I. Consolidated Technical Specification body

This v1.7 consolidates the prior technical body and all controlled correction passes. Historical version references are non-normative unless explicitly marked as current in the exact baseline table.

Паспорт документа

Поле	Значення
Ідентифікатор	LINGVESSA-TECH-SPEC
Версія / дата	1.7 / 09.09.2026
Статус	OWNER-LOCKED IMPLEMENTATION BASELINE; owner lock 10.09.2026; P0 authorized.
Замінює	Technical Specification v1.6; previous versions remain historical.
Нормативна основа	Formal Specification v1.7; Technical Specification v1.7; Technical Design v1.7; Curriculum Blueprint A1 v1.6; Reference Module Design v1.6; AI Integration Specification v1.4; Implementation Plan v1.4 (execution order only).
Аудиторія	Власник продукту, frontend/backend, content/admin tooling, DevOps/SRE, security, QA, linguistic/pedagogical reviewers.
Мови	UI/explanations: uk-UA; target: en-GB default, en-US selectable, cs- CZ.
Repository	Private Git repository; GitHub Private is development default.
Provider policy	Provider-neutral contracts; free/local-first AI; paid AI disabled by default.

Нормативні слова

MUST - release-blocking; SHOULD - default with documented rationale for deviation; MAY - compatible extension. У конфлікті Formal v1.7 має продуктову перевагу, а Technical Design v1.7 деталізує implementation contract. AI Integration Specification v1.4 є нормативним для AI-specific behavior.

Що змінено у v1.7 correction pass

Зміна	Результат v1.7

Product identity	Уся технічна документація та namespace використовують робочу назву Lingvessa.
Mastery	Введено two-level model: 6 global skill projections + knowledge-item facets; окремі tables/projections і evidence mapping.
Pronunciation	Usable speech -> learner-facing correct/try again; low confidence/mic denial/provider failure -> neutral speech unavailable. Raw score не _ показується.
AI Layer	Додано AI Orchestrator, provider adapters, learner AI та admin AI, permissions, quotas, fallback, audit і non-dependency guarantee.
AI administration	Manual / AI Assist / AI Auto. AI Auto allowlist-based; publish/rollback/delete/role/billing заборонені автономно у v1.
Cost policy	Paid AI = OFF by default; initial paid AI budget = 0. Free/local-first не скасовує privacy/QA.
Implementation path	Прибрано обов'язковий one-folder prototype. P1 = production-oriented bilingual reference implementation.
Deployment	Зафіксовано development default: GitHub Private + Northflank Developer Sandbox + Cloudflare R2; production provider remains ADR.
Professional tracks	Architecture support in v1; full Digital Management/PPC content is non- blocking for core launch.

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ЧАСТИНА A. РІШЕННЯ ТА СИСТЕМНА АРХІТЕКТУРА

1. Призначення, межі та precedence

Цей документ перетворює Formal Specification v1.7 на реалізовний software blueprint. Lingvessa будується як одна multilingual platform для English і Čeština з українським UI, спільними engines і language-specific configuration. Core learner

path не залежить від доступності AI, live media generation або зовнішнього speech provider.

У межах technical baseline v1	Поза межами / окреме рішення
Responsive Web/PWA; learner UI; Admin/CMS; identity; curriculum; activities; attempts; two-level mastery; review; collections; audio/speech; media pipeline; optional AI layer; analytics/operations.	Native mobile apps, payments, social network, marketplace, live tutors, PDF/textbook import, official CEFR certification.
Architecture/schemas for A1-C1 English + Czech; module-by-module publication after QA.	Обіцянка одночасної повної A1-C1 бібліотеки без capacity/editorial evidence.
Provider-neutral AI/TTS/STT/image/email integrations.	Будь-який vendor як незамінна domain dependency.

ID	Вимога	Пріоритет	Критерій перевірки
TS-GOV-001	Formal v1.7 requirement IDs мають trace до code/schema/test/ADR.	MUST	Release checklist не містить orphan MUST.
TS-GOV-002	Implementation convenience не змінює product scope без approved decision.	MUST	ADR + spec update.
TS-GOV-003	AI/provider/region/payment/age decisions, що не зафіксовані, лишаються open.	MUST	No hidden production default.
TS-GOV-004	Technical minor version змінюється після incompatible contract/architecture change.	MUST	Change log + migration note.

2. Архітектурні цілі та принципи

- Multilingual by construction: language profile, content, cache, evidence, review, media, AI context і analytics мають explicit

language/variety scope.

- Modular monolith first: одна application boundary з enforceable modules; workers додаються для довгих/cost-heavy jobs.
- Content-driven: React/UI не містить конкретного lesson content або course order.
- Server-authoritative learning: client не визначає correct/mastery/due_at/publish.
- Immutable evidence/publications: історія attempts і published snapshots не переписується мовчки.
- Provider-neutral ports: TTS/STT/image/AI/email/storage integrations behind adapters.
- AI is optional capability layer: core learning remains fully functional when AI disabled.
- Progressive delivery: bilingual reference slice before bulk content and before microservices.

ID	Вимога	Пріоритет	Критерій перевірки
TS-ARCH-001	English і Čeština використовують однакові application services та activity contracts.	MUST	No language-specific engine fork.
TS-ARCH-002	Core lesson/attempt path не залежить від AI/media-generation availability.	MUST	AI/provider outage fault test.
TS-ARCH-003	PostgreSQL authoritative; cache/analytics/AI memory non-authoritative.	MUST	Cache/AI reset does not alter progress.
TS-ARCH-004	Cross-module writes only through declared application ports/events.	MUST	Architecture lint.
TS-ARCH-005	Microservice extraction only after measured need + ADR.	SHOULD	ADR with data ownership/failure/migration.

3. System context i runtime topology

Runtime topology

Web/PWA + Admin/CMS -> Application API (NestJS modular monolith) -> PostgreSQL. Workers consume outbox/jobs. Redis-compatible queue/cache MAY be introduced by stage. Object storage/CDN hosts immutable media. Provider adapters isolate AI/TTS/STT/image/email. Core synchronous learner flow bypasses AI/media generation.

Компонент	Відповідальність	Не повинен робити
Web/PWA	Learner UX, local queue, media playback/record, AI UI surfaces.	Authoritative scoring/mastery; secrets.
Admin/CMS	Draft/edit/preview/review/publish/rollback; AI settings and review.	Bypass validation/approval or execute hidden autonomous publish.
Application API	AuthZ, domain commands/queries, answer evaluation, AI orchestration boundary, transactions, OpenAPI, outbox.	Long generation/transcode/model jobs in request transaction.
Workers	Media/AI jobs, validation, notifications, cleanup, projections.	Mutate published content outside workflow.
PostgreSQL	Users, content, attempts/evidence, skill/facet mastery, review, AI audit/config, publications, outbox.	Large media bytes.
Redis/queue	Optional cache/locks/queue/rate limits.	Sole progress/publication store.
Object storage/CDN	Versioned images/audio/assets.	Canonical provider URLs.
Providers	AI/TTS/STT/image/email via adapters.	Receive unnecessary personal data or domain authority.

4. Технологічний baseline і version policy

Перевірено 04.09.2026

Node.js 24 is LTS; Node.js recommends production use of Active/Maintenance LTS. PostgreSQL 18.6 is the current minor for PostgreSQL 18. Framework majors are pinned only after compatibility spike and lockfile review.

Шар	Baseline	Правило
Runtime/tooling	Node.js 24 LTS; TypeScript strict; pnpm workspaces; lockfile.	Exact versions pinned in repository.
Web	React + Next.js App Router + design tokens/PWA.	Original visual system; accessible primitives.
API	NestJS modular monolith; REST JSON; OpenAPI 3.1.2.	/api/v1; Fastify optional after benchmark.
Data	PostgreSQL 18.6/current minor; Prisma migrations/client + SQL where required.	JSONB only for versioned payloads/metadata.
Schema	JSON Schema 2020-12 + generated TS/runtime validation.	One canonical schema for authoring/API/tests.
Queue/cache	Adapter interface; Postgres/outbox first, Redis-compatible/BullMQ when measured need.	Durable state remains PostgreSQL/object storage.
AI	Provider-agnostic AI Orchestrator + adapters; free/local-first.	Paid provider disabled unless budget/config explicitly enables.
Media	S3-compatible object storage + CDN; FFmpeg/image tools.	Immutable keys/checksums.
Testing	Unit/property + Testcontainers + Playwright + axe/manual a11y.	English+Czech fixtures for core activities.
Delivery	OCI containers, CI/CD, secret manager, IaC where useful.	Repeatable local/preview/staging/prod.

ID	Вимога	Пріоритет	Критерій перевірки
TS-TECH-001	Runtime/framework/package manager pinned; reproducible clean checkout.	MUST	CI install/build.
TS-TECH-002	Node production major must remain supported LTS.	MUST	Lifecycle job.
TS-TECH-003	PostgreSQL runs supported major/current security minor.	MUST	Patch cadence.
TS-TECH-004	Major upgrade requires	MUST	Upgrade PR gate.
ID	Вимога	Пріоритет	Критерій перевірки
	compatibility/migration/E2E/rollback evidence.		
TS-TECH-005	AI SDK types may not leak into domain/content schema.	MUST	Forbidden import / adapter contract.

5. Monorepo, module boundaries i dependency rules

/apps/web learner PWA

/apps/admin CMS/Admin

/apps/api NestJS HTTP application

/apps/worker long-running jobs when separated

/packages/domain

/packages/application

/packages/activity-schemas

/packages/lesson-engine

/packages/ai-core

/packages/provider-adapters # subfolders: ai/, speech/, tts/, image/, email/, storage/

/packages/api-client

/packages/ui

/packages/content-tools

/content/english /content/czech /prisma /infra /docs/adr

Module	Owns	May call
identity	users, sessions, roles, consents	audit/notifications ports
curriculum/content	languages, courses, lessons, activity versions, publications	knowledge/media/audit
learning	lesson runs, attempts, evidence, mistakes	content read model, mastery/review
mastery/review	skill/facet states, schedules, sessions	immutable evidence/content metadata
ai/orchestration	AI policy, routing, quotas, prompt/version refs, learner/admin AI results	content read models, mastery summary, provider adapters; no direct publish
media/speech	assets, voices, speech results, provider adapters	content requirements, audit
admin/workflow	approvals, publish/rollback, AI permissions/config	all modules through application ports
analytics/ops	outbox consumers/projections/metrics	read events only

ID	Вимога	Пріоритет	Критерій перевірки
TS-MOD-001	Domain package has no framework/provider SDK imports.	MUST	Dependency lint.
TS-MOD-002	Provider SDK only in infrastructure adapter package.	MUST	Forbidden import rule.
TS-MOD-003	AI orchestrator cannot write publication/role/billing tables directly.	MUST	Architecture/security tests.
TS-MOD-004	Cross-module write via port/use case/domain event.	MUST	Integration tests.

6. Environments, configuration i deployment

Environment	Призначення	Правила
local	Synthetic fixtures/local services.	No production secrets/data; one-command bootstrap.
preview	Per-PR UI/API preview.	Protected access; auto-expiry; no production providers.
staging	Production-like topology + sandbox providers.	Migration/load/security/E2E/publish/rollback/AI fallback drills.
production	Real users/content.	MFA admin; approved providers/regions; backups/SLO/incident ownership.

Development deployment default

Source code: GitHub Private. For low-cost development/reference implementation: Northflank Developer Sandbox MAY host two services plus one database addon; API and worker may be co-located in one service only for development. Cloudflare R2 Standard is the preferred development object storage. This is not a production lock-in: containers + PostgreSQL + S3-compatible storage remain portable.

ID	Вимога	Пріоритет	Критерій перевірки
TS-ENV-001	All environments share one typed config schema.	MUST	Startup config tests.
TS-ENV-002	Secrets absent from repository/client/logs.	MUST	Secret scans.
TS-ENV-003	Production deploy references immutable artifact + migration set.	MUST	Release record.
TS-ENV-004	AI paid spend defaults to zero and requires explicit production config/approval.	MUST	Config test + audit.
TS-ENV-005	Provider/AI feature flags fail safe and do not replace authorization.	MUST	Negative tests.

7. Repository, source control і release provenance

- All source code, schemas, migrations, tests, documentation, prompts/templates and manifests are versioned in a private Git

repository.

- Large generated images/audio are not committed by default; repository stores manifests, checksums and generation

metadata while bytes live in object storage.

- Protected main branch; reviewed pull requests; signed commits/releases where feasible; tags bind code + schema + content

compatibility.

- AI prompts/policies that affect learner/admin behavior are versioned like code and referenced by execution records.

ID	Вимога	Пріоритет	Критерій перевірки
TS-REPO-001	Git repository is the source of truth for code/schema/migrations/docs.	MUST	Clean checkout rebuilds artifacts.
TS-REPO-002	No production secret or large generated-media corpus in Git history.	MUST	Secret/size checks.
TS-REPO-003	Release provenance links commit, image digest, DB/content/AI policy versions.	MUST	Release reconstruction drill.

ЧАСТИНА B. ДАНІ, КОНТЕНТ І LEARNING ENGINES

8. Data conventions і language isolation

Конвенція	Рішення
IDs	Opaque UUID; UUIDv7 preferred when supported. Slug/title/order never identity.
Time	PostgreSQL timestamptz UTC; IANA user timezone; review boundary server-side.
Languages	BCP47 uk-UA, en-GB, en-US, cs-CZ; target language separate from UI locale.
Unicode	UTF-8; NFC comparison policy by field; no global Czech diacritic stripping.
Scores	Internal 0..1 estimates + uncertainty/version; UI is derived.
AI context	No uncontrolled full-history dump; purpose-limited context bundle with explicit language/profile/task scope.
JSONB	Versioned activity/provider/AI metadata only; ownership/status/due at _ remain relational.

Language isolation invariant

Every learner state key begins with user_id + language_profile_id. knowledge/content/course references are validated against that same target language. AI prompts and conversation memory are also scoped to the active language profile; switching language never cross-feeds progress or memory.

ID	Вимога	Пріоритет	Критерій перевірки
TS-DATA-001	Unique language profile per user/language.	MUST	Duplicate rejected; switch preserves both states.
TS-DATA-002	DB/service guard blocks cross- language attempt/mastery/review/AI memory/collection.	MUST	Negative mixed-data tests.

TS-DATA-003	User-scoped reads/writes check ownership server-side.	MUST	IDOR tests.
TS-DATA-004	Unicode round-trip preserves Ukrainian/English/Czech.	MUST	Checksum fixture.

9. Core relational model

Area	Основні tables / fields	Ключові constraints
Identity	users, credentials, sessions, roles, user roles, _ consents	normalized identity; audited role change
Profiles	language profiles, user settings, _ _ notification preferences _	unique user+language
Curriculum	languages, variants, courses, levels, modules, lessons, activities/items	language lineage; stable IDs
Knowledge	outcomes, knowledge items, knowledge edges, _ _ item knowledge map _ _	skill dimension + facet mapping _
Learning	lesson runs, activity attempts, evidence events, _ _ _ mistake instances _	immutable attempts/evidence; content version retained
Mastery	skill mastery states, knowledge mastery states _ _ _ _	global skill key vs knowledge+facet key
Review	review schedules, review sessions/items _ _	knowledge+facet scheduler version; language isolated
AI	ai_provider_profiles, ai_model_profiles, ai_policies, ai_prompt_templates, ai_runs, ai_sessions, ai_messages, ai_artifacts, ai_usage_buckets, ai_tool_actions	policy/prompt/provider/model versions; permissions and budget eligibility in ai_policies; usage in ai_usage_buckets; no secrets in audit payload
Collections	collections, entries	owner+language
Media	requirements, jobs, assets, asset versions, voices, _ characters, speech results _	version/checksum/provenance
Publication	content versions, publications, entries, approvals _	immutable snapshot
Operations	outbox, audit, idempotency, feature flags	append-only/redacted where required

ID	Вимога	Пріоритет	Критерій перевірки
TS-DATA-005	Critical uniqueness/referential integrity enforced in DB.	MUST	Concurrent conflict tests.
TS-DATA-006	Attempts/evidence/audit/outbox immutable except documented redaction/tombstone.	MUST	No generic update endpoint.
TS-DATA-007	AI execution records store policy/prompt/model/provider versions without raw secrets.	MUST	Audit/privacy fixture.
TS-DATA-008	Cascade delete only where domain lifecycle explicitly allows.	MUST	Schema review.

10. Curriculum, knowledge graph і publication model

Curriculum defines ordered learning path; knowledge graph defines prerequisites, contrasts, components, variants, reuse and

co-occurrence. Published lessons reference immutable content_version + publication manifest. Knowledge IDs survive surface

edits so mastery is not lost through wording changes.

Entity	Required fields
KnowledgeItem	id, language, variety scope, type, canonical form/sense, CEFR scope, status, _ schema version
KnowledgeEdge	from/to, edge type, direction, weight, version, review status, provenance _ _
Outcome	language, CEFR level, can-do, source reference, status/version
ContentVersion	entity, version, schema version, payload, checksum, status, author/time
Publication	course/channel/version, manifest checksum, status, previous publication
Approval	target_content_version_id, target_checksum, gate_type, reviewer_id, decision, decided_at, notes; checksum-sensitive. Documentation baseline approval is separate governance metadata.

ID	Вимога	Пріоритет	Критерій перевірки
TS-CONT-001	Published content/asset versions immutable.	MUST	Checksum and overwrite tests.
TS-CONT-002	Publication manifest sufficient for	MUST	Manifest restore.

	replay/rollback.		
TS-CONT-003	Assessable item has outcome + knowledge + skill/facet evidence mapping.	MUST	Preflight blocks missing mapping.
TS-CONT-004	Knowledge split/merge/deprecate includes migration impact report.	MUST	No silent mastery loss.
TS-CONT-005	AI-generated draft has provenance and cannot bypass human/validation gates.	MUST	Workflow test.

11. Activity contract i registry

Contract block	Required fields
Identity/scope	id, schema version, type, language, variety, CEFR, content version _ _
Presentation	instruction key, prompt, stimulus, media refs, accessibility, speaker/lang _ _ segments
Response	response schema, answer policy id, accepted answers/references, _ _ _ distractors
Feedback	feedback rules, hint/reveal/remediation policies
Learning	knowledge refs, outcome refs, skill dimension, facet code, evidence _ _ _ _ weight/difficulty
Runtime	randomization seed/order, time policy, analytics tags, feature requirements, AI enhancement optionality

ID	Вимога	Пріоритет	Критерій перевірки
TS-ACT-001	Registry maps type+schema major to renderer/evaluator/fixtures.	MUST	Contract suite.
TS-ACT-002	Educational content not hardcoded in renderer.	MUST	Fixture swap.
TS-ACT-003	Unknown schema/type recoverable; no corrupt attempt.	MUST	E2E error boundary.
TS-ACT-004	AI enhancement cannot be required unless activity explicitly marks an optional AI path with deterministic non-AI fallback.	MUST	AI-off E2E.

12. Answer evaluation i exercise state machine

Scored activity

not_started -> active -> submitted -> correct | incorrect | assisted. incorrect -> feedback -> retry | simpler scaffold | reveal. Next is allowed only from completed. completed for scored items is reachable only after correct or explicit reveal; skip/timeout/network failure is never success.

Binary speech with safe fallback

idle -> permission_request -> recording -> processing -> speech_correct | speech_incorrect | speech_unavailable. Usable, sufficiently confident speech maps to learner-facing correct / try again. speech_incorrect MAY create bounded negative evidence only for targeted Pronunciation facet. speech_unavailable covers denial, absent mic, timeout, low confidence and provider failure; it creates no mastery update and never blocks lesson/recap.

ID	Вимога	Пріоритет	Критерій перевірки
TS-EVAL-001	Server selects versioned answer policy; client correct flag ignored.	MUST	Tamper test.
TS-EVAL-002	Reveal/answer-giving hint sets assisted=true and caps productive evidence.	MUST	Independent vs assisted test.
TS-EVAL-003	Czech diacritics and English variety behavior explicit per policy.	MUST	Language fixtures.
TS-EVAL-004	Speech result uses versioned language/variant pronunciation policy; raw provider score hidden from learner.	MUST	Speech fixtures.
TS-EVAL-005	AI-generated feedback does not override deterministic correct answer for closed-form scored activities.	MUST	Provider disagreement test.

13. Attempt command, transaction і idempotency

Client generates opaque attempt_id and sends pinned content_version, lesson_run_id, item_id, response, presentation seed/order, assistance metadata and client sequence. Server validates ownership/scope/version, evaluates, then atomically writes attempt + evidence + lesson position + mistake/mastery/review effects + outbox event. Provider/analytics work happens

after commit.

ID	Вимога	Пріоритет	Критерій перевірки
TS-ATT-001	UNIQUE(user id, attempt id) gives _ _ exactly-once credit under at-least- once delivery.	MUST	Concurrent retry test.
TS-ATT-002	Same ID + same semantic payload returns stored result; different payload -> conflict.	MUST	Retry/conflict tests.
TS-ATT-003	Attempt/evidence/learning state/outbox commit atomically.	MUST	Fault injection.
TS-ATT-004	Attempt references exact content/evaluator/mastery/scheduler versions.	MUST	Historical replay.
TS-ATT-005	AI/provider failure after core commit	MUST	Consumer failure test.
ID	**Вимога**	**Пріоритет**	**Критерій перевірки**
	cannot rollback accepted progress.		

14. Two-level evidence, mastery і mistake remediation

Mastery model

Level 1 - global skill projections: Vocabulary, Grammar, Listening, Speaking, Writing, Pronunciation. Level 2 - knowledge-item facets: recognition, meaning, recall/production, spelling, listening recognition, pronunciation, context_use, collocation,

grammar_form, word_order, morphology, register та language-specific facets. A single task may emit evidence to one or more explicitly mapped targets.

Object	Canonical content
EvidenceEvent	attempt, profile, knowledge item(optional), skill dimension(optional), _ facet code(optional), result class, assistance, difficulty, weight, _ model/evaluator/content versions, time
SkillMasteryState	profile + skill + model version -> estimate, uncertainty, counts, lapses, last _ evidence
KnowledgeMasteryState	profile + knowledge + facet + model version -> estimate, uncertainty, counts, _ lapses, last evidence
MistakeInstance	attempt/item/knowledge/facet, error type, first/last seen, unresolved/remediated/scheduled
Skill aggregate	Derived CEFR/course view; never primary evidence store

ID	Вимога	Пріоритет	Критерій перевірки
TS-MAST-001	Evidence immutable; mastery projections recomputable by model version.	MUST	Recompute fixture.
TS-MAST-002	One knowledge item surface across lessons updates same facet state.	MUST	Duplicate-surface test.
TS-MAST-003	Evidence weight depends on operation, difficulty, hints/reveal, attempts/freshness.	MUST	Property tests.
TS-MAST-004	Correct remediation closes mistake but retains lapse and shorter review effect.	MUST	State/scheduler test.
TS-MAST-005	AI learner output may update mastery only through an approved rubric/evaluator that emits structured evidence; model prose alone is not evidence.	MUST	AI evidence gate test.

15. Review scheduler i session builder

Stage	Rule
Candidate	due first; unresolved mistakes; weak/high uncertainty; user-selected topic/collection; limited recycle/new
Deduplication	group by knowledge+facet and normalized surface; avoid near duplicates
Ordering	interleave skills without chaos; language hard filter; session cap/ETA
Scheduling	versioned stability/difficulty/interval state; incorrect/reveal/hint/skip differ
AI enhancement	MAY generate additional practice surface from approved knowledge targets; generated item must validate before learner exposure and may be ephemeral with provenance

ID	Вимога	Пріоритет	Критерій перевірки
TS-REV-001	Review query starts from language profile id. _ _	MUST	Mixed dataset.
TS-REV-002	Scheduler deterministic for same state/result/time + version.	MUST	Golden tests.
TS-REV-003	Intervals configurable; no hardcoded UI schedule.	MUST	Policy config test.
TS-REV-004	AI-generated review item cannot change due state unless submitted through normal attempt/evidence path.	MUST	AI review test.

16. Lesson run, resume i offline synchronization

- Server stores pinned lesson_run content/publication version, state, position and sequence.
- Client stores non-sensitive presentation state and unsent attempts in IndexedDB; no raw mic audio.
- Reconnect preserves original client sequence and attempt IDs.
- Publication change does not mutate active run; compatibility policy controls migration.
- AI conversation sessions are not required for offline completion; if unavailable, deterministic fallback is used.

ID	Вимога	Пріоритет	Критерій перевірки
TS-SYNC-001	Reload/app restart resumes authoritative run + reconciled queue.	MUST	E2E every core state.
TS-SYNC-002	Offline queue has retention cap and excludes secrets/raw audio.	MUST	Storage inspection.
TS-SYNC-003	Reconnect uses idempotent attempt protocol.	MUST	Network flap test.
TS-SYNC-004	AI-unavailable/offline state never fabricates AI result or blocks core lesson.	MUST	Offline AI-off E2E.

ЧАСТИНА C. API, JOBS, AI, MEDIA ТА INTEGRATIONS

17. API conventions i compatibility

Сфера	Contract
Base	HTTPS JSON REST /api/v1; OpenAPI 3.1.2 generated/validated.
Identity	Authenticated user from session; language profile explicit for learner _ resources.
Validation	Runtime schema; invalid enum/version/language/ownership rejected at boundary.
Errors	application/problem+json-compatible: stable code, safe detail, correlation id, _ field errors.
Concurrency	ETag/If-Match or version field for mutable settings/drafts.
Idempotency	attempt id domain key; Idempotency-Key for destructive admin/job/AI _ commands.
Caching	Private auth responses no-store unless safe; versioned public content ETag/immutable caching.
AI	Streaming endpoint optional; no provider-native response shape exposed to clients.

ID	Вимога	Пріоритет	Критерій перевірки
TS-API-001	OpenAPI checked against implementation and typed client.	MUST	Contract diff.
TS-API-002	Errors exclude stack/SQL/provider	MUST	Negative tests.

	secrets.		
TS-API-003	Mutating endpoints validate auth/AuthZ/CSRF/body/runtime schema.	MUST	Security matrix.
TS-API-004	Client cannot set score/due/role/publish/AI permission/budget.	MUST	Tampering tests.
TS-API-005	Breaking contract versioned with migration/deprecation plan.	MUST	OpenAPI diff gate.

18. Learner API surface

Method/path	Purpose	Critical contract
POST /auth/register|login|logout	Identity/session	Rate limit; age/jurisdiction configurable before public registration
GET/PATCH /me	Profile	Versioned settings; no secrets
GET /me/language-profiles	English/Czech states	Independent position/variety/due counts
PUT /me/active-language	Switch target context	No progress mutation
GET /catalog?language=	Course map	Published projection only
POST/GET /lesson-runs	Start/resume lesson	Pinned version
POST /lesson-runs/{id}/attempts	Evaluate/persist attempt	Idempotent + transaction
POST /review-sessions	Build queue	language/mode/topic/collection/cap
GET /mastery/summary	Derived global skills + optional facet detail	Uncertainty-aware
POST /ai/conversations	Optional AI role-play	Feature flag, quota, language scope, fallback
POST /ai/explain	Optional personalized explanation	No mutation of canonical curriculum
POST /ai/writing-feedback	Optional structured feedback	Rubric/version/provenance; no direct mastery unless approved evaluator
CRUD /collections	Personal sets	Owner+language
GET/PATCH /settings	Audio/auto-advance/AI learner opt-in	Server source of truth
POST /data-export; DELETE /account	Privacy	Reauth + async status + audit
POST /speech/jobs	Create/reuse speech job	Idempotency-Key + attempt_id; owner/profile/item/content pinned; returns job_id + transient upload_url.
POST /speech/jobs/{job_id}/submit	Queue speech scoring	Owner scoped; idempotent; upload checksum/TTL validated.
GET /speech/jobs/{job_id}	Poll speech status/result	Owner scoped; terminal result settles attempt once.
POST /speech/jobs/{job_id}/cancel	Cancel speech scoring	Idempotent; settle speech_unavailable(reason=user_cancelled); late callbacks ignored.

19. Admin, content i publication API

Capability	Commands/queries	Gate
Drafting	create content/version, validate, preview	editor role, schema/lint
Review	linguistic/pedagogical/technical decisions	review separation policy
Media	requirements/jobs/approve/reject/regenerate	provenance/cost/quality
AI config	provider/model priority, mode, quotas, permissions, budgets	admin + MFA/reauth for budget/provider activation
AI draft	generate lesson/media/prompt suggestions	AI Assist/Auto allowlist; provenance; dry-run
Publication	preflight, publish snapshot, rollback	mandatory approvals + checksums + admin/MFA
Operations	jobs/audit/provider health/cost/validation	no secrets/raw audio

ID	Вимога	Пріоритет	Критерій перевірки
TS-ADM-001	Draft save and approval are separate commands/states.	MUST	Workflow transition tests.
TS-ADM-002	Publish preflight repeatable against exact checksums.	MUST	Mutation invalidates approval.
TS-ADM-003	AI generator/service role cannot publish/rollback/delete/change role/billing.	MUST	Permission matrix.

TS-ADM-004	AI provider/budget/mode changes are audited.	MUST	Audit event includes old/new safe values.
TS-ADM-005	Bulk/AI generation supports dry-run, bounded batch, progress and partial-failure report.	MUST	Retry idempotency.

20. Jobs, queues, outbox i callbacks

Job class	Examples	Default behavior
Content	validate, coverage, manifest, publish preflight	deterministic checksum; no publish without approval
Media	image/audio generate, transcode, QA	quota, timeout, idempotent ingest, bounded retry
AI	conversation summary, draft generation, explanation/writing feedback async, content QA	provider routing, budget, timeout, schema-validated output, policy version
Speech	async scoring	short-lived bytes; cleanup
Privacy	export/deletion/retention	high priority, resumable, audited
Projection	analytics/search/coverage	rebuildable; core commit unaffected

ID	Вимога	Пріоритет	Критерій перевірки
TS-JOB-001	Domain transaction writes outbox atomically.	MUST	Crash delivery test.
TS-JOB-002	Consumer idempotent by event/job key.	MUST	Duplicate delivery test.
TS-JOB-003	Retries bounded with backoff+jitter and terminal classification.	MUST	Permanent error test.
TS-JOB-004	AI/media job records provider/model/version/cost tokens/latency without sensitive payload.	MUST	Telemetry schema test.
TS-JOB-005	Provider callbacks verify signature/timestamp/job/provider/repl ay window.	MUST	Forged callback test.

21. AI capability layer

Core guarantee

AI is optional. Disabling every AI provider and AI feature flag MUST leave onboarding, catalog, lessons, deterministic scoring, Review, collections, progress, audio playback and account controls operational.

Layer	Responsibility
AI Orchestrator	Task classification, policy lookup, provider routing/fallback, schema validation, quotas/budget, audit.
Provider adapters	Gemini/Groq/OpenRouter/local/Ollama or future providers; internal normalized request/result.
Learner AI	Conversation/role-play, personalized explanation, writing feedback, adaptive hints, extra practice.
Admin AI	Draft content/media specs, QA suggestions, anomaly analysis, controlled automation.
Policy/permissions	Manual / AI Assist / AI Auto; action allowlist by risk and role.
Cost control	Paid budget 0 by default; per-provider/model quotas; hard stop before paid overage.
Fallback	Provider failure -> another approved provider/local model -> deterministic non- AI fallback where task allows.

Mode	Behavior
Manual	AI disabled or suggestion-only on explicit request; no automatic changes.
AI Assist	AI produces proposed result/diff; human confirms before mutation/publication-sensitive use.
AI Auto	Only explicitly allowlisted low-risk operations; all executions audited. No autonomous publish, rollback, account deletion, role change, billing/payment, certification decision in v1.

ID	Вимога	Пріоритет	Критерій перевірки
TS-AI-001	AI provider SDKs remain behind adapters; no provider-native IDs/shapes in domain schema.	MUST	Adapter substitution contract.
TS-AI-002	AI runs pin task, policy, prompt/template, provider, model and output schema version.	MUST	Replay/audit record.
TS-AI-003	Paid AI is disabled by default and cannot incur cost without explicit enabled budget.	MUST	Zero-budget test.

TS-AI-004	AI Auto permissions are allowlist- based and deny sensitive actions.	MUST	Negative permission matrix.
TS-AI-005	Model output must pass JSON/runtime validation and applicable linguistic/pedagogical/content rules before use.	MUST	Malformed/hallucinated output tests.
TS-AI-006	AI learner feedback cannot claim official CEFR certification or medical/legal/other high-stakes authority.	MUST	Prompt/policy tests.
TS-AI-007	AI context is data-minimized and language-profile scoped.	MUST	Privacy payload audit.
TS-AI-008	Provider outage/quota exhaustion produces user-safe fallback and no core progress loss.	MUST	Fault injection.

22. Media, voices, audio i speech

Object	Required metadata/rule
Asset	stable asset id, language/purpose, provenance/rights, active version _
AssetVersion	object key, checksum, MIME, dimensions/duration, codec/loudness, generation model/spec hash, QA
Object	**Required metadata/rule**
Voice	provider adapter ID, en-GB/en-US/cs-CZ scope, profile/style, licence/consent
Character	versioned visual reference, style profile, voice binding, allowed contexts
SpeechResult	language/variety, engine/model, result class, confidence, error tags, threshold policy; raw recording normally absent

- Normal audio required where lesson contract marks it; slow uses pitch-preserving transform or reviewed slow asset.
- Microphone permission only after explicit user action. UI microphone may pulse while recording; reduced-motion changes

the visual behavior without hiding state.

- Usable speech -> speech_correct/speech_incorrect and learner-facing correct/try again. Auto-advance may fire only after

result is visibly rendered.

- speech_unavailable = denied/absent mic/timeout/low confidence/provider failure; no mastery update and no hard block.
- Real-person voice/face cloning requires explicit lawful basis/consent; default is original/synthetic/licensed assets.

ID	Вимога	Пріоритет	Критерій перевірки
TS-MEDIA-001	Missing/broken/unapproved required asset blocks publish.	MUST	Manifest preflight.
TS-MEDIA-002	Regeneration creates new immutable version/checksum.	MUST	Rollback cache test.
TS-MEDIA-003	Raw mic audio absent from logs/analytics/object storage by default.	MUST	Speech E2E storage audit.
TS-MEDIA-004	speech incorrect affects only targeted _ pronunciation facet and bounded review; speech unavailable affects _ none.	MUST	Mastery fixture.
TS-MEDIA-005	Images/audio generated by system before normal course publication; learner never must source media.	MUST	Published lesson provider-offline test.

23. Object storage, CDN i cache

Layer	Policy	Failure behavior
Object storage	Private origin; immutable asset/version/checksum key; encryption/lifecycle/access logs.	Signed admin operations; published delivery via CDN.
CDN	Immutable path/checksum, correct MIME/ETag/CORS/CSP.	No fallback to provider URL.
API cache	Cache-aside; key includes locale/language/publication/auth scope.	Miss -> DB; purge by version.
Client cache	PWA shell/versioned assets; no secrets.	Clear/upgrade without losing server progress.

Development storage default

Cloudflare R2 Standard is the preferred development object store because it is S3-compatible and has a free monthly tier; production storage remains replaceable by adapter/configuration.

24. Domain events i analytics contracts

Event family	Examples	Required context
Learning	lesson/activity/review result, hint/reveal	pseudonymous profile, language, content/evaluator/mastery/scheduler versions; no raw audio
AI learner	ai conversation turn, ai explanation, _ _ _ ai writing feedback _ _	task/policy/provider/model versions, latency/cost class, safety/fallback result; no unrestricted prompt transcript by default
Content/Admin AI	ai draft created, ai suggestion accepted/rejected, _ _ _ _ ai auto action _ _	actor/mode/action/target/diff ref/policy/provider versions
Operational	request/job/provider latency/error, queue/cache	service/env/version/correlation
Security/audit	login anomaly,	actor/target/action/result/time
Event family	**Examples**	**Required context**
	role/publish/delete/export/provider/budget action	

ID	Вимога	Пріоритет	Критерій перевірки
TS-EVT-001	Events versioned with correlation/causation IDs.	MUST	Schema registry tests.
TS-EVT-002	Learning/AI diagnostics include model/policy/content versions when applicable.	MUST	Support trace.
TS-EVT-003	Analytics allowlist excludes raw audio/tokens/passwords/full free responses by default.	MUST	Payload privacy tests.
TS-EVT-004	Analytics sink failure never blocks accepted attempt.	MUST	Sink outage test.

ЧАСТИНА D. WEB/PWA, LESSON PLAYER TA ADMIN UI

25. Web/PWA architecture

Concern	Рішення
Routing	Next.js App Router; explicit language profile context.
Rendering	Published catalog may SSR/revalidate; LessonPlayer is validated client boundary.
Server state	Generated typed API client/query cache; invalidation by profile/publication keys.
Lesson state	Dedicated finite state reducer; not scattered booleans.
AI state	AI requests separate cancellable/stream state; never conflated with authoritative lesson attempt state.
Offline	IndexedDB attempt queue + selected assets; no full offline promise until accepted.
Design system	Original tokens/primitives; no copied trade dress.
Error boundaries	Safe code + retry/exit/resume; diagnostics include version/correlation.

ID	Вимога	Пріоритет	Критерій перевірки
TS-WEB-001	Active target language explicit and visible.	MUST	Language switch E2E.
TS-WEB-002	No secrets/provider keys in client bundle.	MUST	Bundle scan.
TS-WEB-003	API/activity/AI structured outputs validated before use.	MUST	Malformed fixture.
TS-WEB-004	AI loading/error cannot disable deterministic lesson controls.	MUST	AI timeout UI test.

26. LessonPlayer i ActivityRenderer lifecycle

State	UI responsibility	Persist/side effect
loading	Skeleton/status, no stale controls	Fetch pinned run/manifest
active	One activity, instruction/media/help/progress	Persist seed/local position

submitting	Prevent duplicate action, allow network retry	Same attempt id _
correct/incorrect/assisted	Announce state + feedback	Authoritative server result
speech correct/speech incorrect _ _	Show correct/try again; mic stops pulsing; next/retry policy	Targeted pronunciation evidence
speech unavailable _	Neutral unavailable; retry or continue	No mastery update
ai enhancing _	Optional explanation/conversation/writing overlay	Separate AI execution; no attempt mutation unless explicit structured evaluator path
remediation	Repeat unresolved scored items	Mistake durable
completed	Outcome summary, mistakes, next/review	Server completion
recoverable error _	Retry/reload/exit safely	No fabricated progress

ID	Вимога	Пріоритет	Критерій перевірки
TS-LESS-001	One primary lesson state controls transitions.	MUST	State-machine tests.
TS-LESS-002	Back/Close/Continue paths for error/speech/AI/network states.	MUST	No-dead-end E2E.
TS-LESS-003	Auto-advance only after authoritative allowed completion state.	MUST	Correct/reveal/speech tests.
TS-LESS-004	AI overlay cannot bypass correct-or-reveal or mutate completion state.	MUST	AI interaction state test.

27. Responsive design, accessibility та i18n

Area	Technical requirement
Viewports	Core flow from 320 CSS px; mobile bottom nav; desktop top/context layout.
Zoom/text	200% reflow; long Ukrainian/Czech strings no clipping.
Keyboard	All actions; visible focus; non-drag alternative.
Media	Alt/decorative marker; transcript/captions; status not only sound/color.
Motion	prefers-reduced-motion; microphone pulse replaced by static/text state when needed.
Language	lang attributes by segment: uk/en-GB/en-US/cs.
AI accessibility	Streaming AI output has aria-live strategy without repeated announcements; stop/cancel keyboard accessible.

ID	Вимога	Пріоритет	Критерій перевірки
TS-A11Y-001	Core learner flow targets WCAG 2.2 AA with automated + manual evidence.	MUST	A11y gate.
TS-A11Y-002	Each activity/AI UI has accessible fallback where gesture/audio/stream is essential.	MUST	Contract gate.
TS-A11Y-003	Mic pulse and AI typing animation respect reduced motion.	MUST	Reduced-motion E2E.

28. Client media, microphone та device behavior

- Audio prefetch bounded; next-step candidates only, not full dialogue corpus.
- Playback exposes play/pause/replay/slow; multiple speakers bound to visible labels/avatar/voice_id.
- Record starts after explicit action; mic icon enters recording state and may pulse; device/codec capability detected.
- Recording duration/size bounded; silence/cancel; buffers released after use.
- No raw mic bytes in analytics/error logs/long-lived IndexedDB.
- When speech result is incorrect, UI shows error state and follows retry/continue/auto-advance policy; when unavailable,

neutral continuation is offered.

29. Admin/CMS frontend and AI controls

Screen	Minimum capability
Course map	Language/course/level/module hierarchy, coverage, prerequisites, gaps/duplicates.
Lesson editor	Structured schema, validation pointers, preview, diff.
Review queue	Linguistic/pedagogical/technical approvals; changed-after-approval warning.
Media	Requirements/jobs/output, provenance, versions, playback, approve/reject/regenerate.
AI settings	Mode Manual/Assist/Auto; provider/model priority; free/local/paid status; per- task permissions; budget/quota; fallback; retention/privacy flags.

AI activity log	Execution, task, actor, target, policy/provider/model version, result, cost class, accepted/rejected/auto action; no secret payload.
Publication	Preflight, manifest diff, migration impact, publish/rollback confirmation.
Operations	Jobs, provider health/cost, AI quota, validation failures, audit, safe retry/cancel.

ID	Вимога	Пріоритет	Критерій перевірки
TS-CMS-001	CMS edits draft versions only; published rows immutable.	MUST	Save behavior.
TS-CMS-002	Preview uses same renderer/schema as learner app.	MUST	Renderer contract.
TS-CMS-003	AI suggestion acceptance creates explicit diff/actor record.	MUST	Audit test.
TS-CMS-004	AI Auto action requires allowlisted permission and dry-run where applicable.	MUST	Policy matrix.
TS-CMS-005	Dangerous actions require human admin/MFA/confirmation and audit.	MUST	Publish/rollback/delete/role/budget E2E.

ЧАСТИНА E. SECURITY, PRIVACY, RELIABILITY TA DELIVERY

30. Authentication, sessions i authorization

Control	Baseline
Credentials	Email/password if enabled; Argon2id parameters benchmarked/versioned; safe recovery.
Session	Revocable server-side session in Secure/HttpOnly/SameSite cookie; rotation on privilege change.
CSRF	SameSite plus origin/token appropriate to topology.
RBAC	learner, editor, linguistic reviewer, pedagogical reviewer, administrator, worker/service.
AI permissions	Task/action-level permission evaluated server-side in addition to role.
Privileged access	MFA + reauth for publish/role/delete/export/provider/budget enablement.
Abuse	Rate limits by action cost; AI/speech/generation quotas separated from ordinary attempts.

ID	Вимога	Пріоритет	Критерій перевірки
TS-AUTH-001	All privileged endpoints enforce RBAC + object/action + AI permission server-side.	MUST	Negative matrix.
TS-AUTH-002	Production administrator requires MFA for critical actions.	MUST	MFA E2E.
TS-AUTH-003	Service identities minimum permissions; AI service cannot impersonate admin/learner.	MUST	Policy test.

31. Privacy, consent i data lifecycle

Data class	Examples	Default lifecycle
Account	email, credentials, consent refs	until deletion/required retention
Learning	attempt metadata, evidence, mastery/review	needed for service; minimize raw response
Voice	transient recording + derived result	raw bytes not retained by default
AI learner	task inputs, conversation turns, writing text, generated feedback	purpose-limited; minimize retention; provider- specific data-use policy reviewed
AI admin	draft prompts/specs/output/diffs	product/content record where needed; no secrets/PII unless explicitly approved
Operational	logs/traces/security events	bounded/redacted
Analytics	pseudonymous events	legal basis/consent by jurisdiction
Generated media	prompt/spec/provenance/provider metadata	product asset retention

Public launch blocker

Countries, minors/parental consent, processors/regions, cookies/analytics, exact retention and provider training/data-use settings must be approved before real-user public registration. Free AI APIs do not bypass these reviews.

ID	Вимога	Пріоритет	Критерій перевірки
TS-PRIV-001	Data inventory maps fields/events to purpose, retention, processor, region,	MUST	Approved register.

	deletion.		
TS-PRIV-002	Mic bytes transient by default.	MUST	Storage/provider cleanup audit.
TS-PRIV-003	AI provider receives only task- minimum data and approved retention/training settings.	MUST	Provider payload audit.
TS-PRIV-004	Export/delete propagates to downstream stores/providers according to policy.	MUST	Deletion drill.

32. Application and supply-chain security

- OWASP ASVS 5.0-oriented verification appropriate to threat model.
- TLS, secure headers/CSP, output encoding, parameterized queries, strict validation, safe media processing.
- Dependency/container/IaC scanning, lockfile review, secrets scan, SBOM/provenance where supported.
- SSRF protection for provider URLs/callbacks; signed webhook verification; outbound allowlist where practical.
- Prompt/provider risks included in threat model: prompt injection from admin content, malicious provider output, tool/action

escalation, data exfiltration, model confusion.

- AI output is untrusted input and schema/safety/authorization validated before any side effect.

ID	Вимога	Пріоритет	Критерій перевірки
TS-SEC-001	Threat model updated for AI/provider/voice/content flows.	MUST	Control matrix.
TS-SEC-002	Critical/high vulnerabilities block release unless time-bounded approved exception.	MUST	CI gate.
TS-SEC-003	AI cannot invoke arbitrary tools/actions outside explicit server allowlist.	MUST	Prompt-injection/action tests.
TS-SEC-004	Provider outputs/uploads treated as untrusted.	MUST	Malformed/polyglot/schema tests.

33. Performance, capacity i budgets

Metric	Initial target	Measurement
Core Web Vitals	p75 LCP <=2.5s, INP <=200ms, CLS <=0.1	RUM + lab
API reads	p95 <=400ms normal cached in-region	server metric
Attempt submit	p95 <=700ms excluding async external processing	E2E API span
Availability	99.5% monthly learner core first production stage	SLI
AI latency	Non-blocking target by task; streaming UI when appropriate; timeout per policy	provider/orchestrator spans
AI budget	Paid AI hard budget = 0 by default; configured quotas for free/local providers	budget ledger + provider usage
Bundle/media	Budgets set after bilingual reference slice	CI + RUM

ID	Вимога	Пріоритет	Критерій перевірки
TS-PERF-001	Reference slice benchmark before scale rollout.	MUST	Recorded benchmark.
TS-PERF-002	Provider call never holds core DB transaction open.	MUST	Trace review.
TS-PERF-003	AI/speech/generation rate/cost limits separate from ordinary attempts.	MUST	Limiter load test.
TS-PERF-004	AI timeout/circuit breaker returns deterministic fallback within bounded wait.	MUST	Fault test.

34. Reliability, backups i disaster recovery

Area	Baseline
Database	Managed PostgreSQL as budget allows; encrypted backups/PITR; first-stage RPO <=24h, RTO <=8h.
Content/media	Publication manifests + critical metadata backed up; object versioning/lifecycle.
AI	Provider/config outage does not affect core; AI policies/prompts/audit backed up as code/data.
Cache/queue	Cache disposable; jobs reconciled from outbox/durable state.

Area	Baseline
Restore	Quarterly isolated restore verifying accounts, publications, attempts, mastery/review, AI configs and representative assets.
Incident	Severity/owner/communication/rollback/disable controls/runbooks.

35. Observability, SLI i alerting

Signal	Required attributes	Examples
Logs	timestamp, service/env/version, correlation, safe code, actor class	auth/publish/job/AI terminal; no raw answer/token/audio
Traces	request/job/AI provider propagation, DB/provider spans	attempt transaction, AI routing/fallback
Metrics	bounded labels/histograms/counters	latency/errors, queue age, provider failure/cost, AI fallback/quota
Learning quality	versioned aggregate events	item ambiguity, hint/reveal, review backlog, speech correct/incorrect/unavailable
AI quality	accept/reject rate, schema failure, fallback, hallucination/quality flags	task/provider/model/policy version aggregates
Alerts	actionable threshold + owner/runbook	DB saturation, queue age, deletion stuck, provider cost anomaly

36. CI/CD, database/content/AI policy migrations i release

Pipeline stage	Required checks/output
Pre-merge	format/lint/typecheck/unit/schema/content/architecture/secret/dependency/AI policy lint
Integration	PostgreSQL/queue/object/provider adapter tests, migrations, outbox/jobs, OpenAPI/JSON Schema
UI/E2E	English+Czech core, AI on/off, pronunciation, viewports, a11y, offline/retry/version skew
Artifact	Immutable build/container, generated schemas/client, provenance/SBOM where supported
Staging	Migrate, smoke/load/security, publish/rollback, AI fallback/budget zero, observability, restore drill subset
Production	Expand migration, rolling/canary, smoke/SLI; contract cleanup later

ID	Вимога	Пріоритет	Критерій перевірки
TS-CICD-001	Production change only through reviewed pipeline; break-glass audited.	MUST	Access review.
TS-CICD-002	DB/content/AI policy migrations tested from current production baseline.	MUST	Staging rehearsal.
TS-CICD-003	Content schema migration includes English+Czech old/new fixtures.	MUST	Compatibility tests.
TS-CICD-004	AI prompt/policy major change has regression, quality and rollback evidence.	MUST	AI release gate.

ЧАСТИНА F. РЕАЛІЗАЦІЯ, ПРИЙМАННЯ ТА РІШЕННЯ

37. Test strategy i release gates

Layer	Scope	Gate
Unit/property	evaluation, state transitions, skill/facet mastery, scheduler, AI policy/routing/budget invariants	Every code change
Schema/content	IDs/refs/language/answers/evidence/media/ provenance/AI output schemas	Every content/schema change
Integration	DB transactions/auth/outbox/jobs/storage/provider adapters/privacy lifecycle	Merge/release
Contract	OpenAPI/JSON Schema/event/AI provider adapters	Breaking change blocked
E2E	onboarding, English/Czech switch, lesson/retry/offline/review/collections/settings/AI on-off/export-delete	Release candidate
Visual/a11y	320px/tablet/desktop, 200% zoom, keyboard/screen reader/motion	UI/core release
Security	ASVS/IDOR/RBAC/CSRF/XSS/secrets/uploads/prompt injection/tool escalation	Public release

Performance/reliability	load/query/bundle, crash/retry/provider outage, restore/rollback	Production readiness
Editorial	linguistic/pedagogical/media/originality + AI-generated content review	Each publication

38. Implementation sequence

Stage	Technical output	Exit gate
Legacy descriptive phase A Contracts	Approved specs, ADR template, monorepo/toolchain, language/activity/content/AI policy schemas, bilingual fixtures, privacy/a11y/threat plan.	Schemas validate English+Czech; AI-off and zero- budget invariants defined.
Legacy descriptive phase A Reference implementation	Production-oriented Web/API/DB skeleton + shared LessonPlayer states using E-A1-M01 and C-A1-M01 representative fixtures.	No dead ends; same engine; pronunciation binary+fallback; no one-folder throwaway gate.
Legacy descriptive phase B Platform foundation	Identity, profiles, catalog, PostgreSQL migrations, environments, CI/CD, GitHub private deployment baseline.	Login + language switch + published manifest on preview/staging.
Legacy descriptive phase C Learning core	Attempt idempotency, immutable evidence, skill/facet mastery, mistakes, resume/offline.	English/Czech E2E + fault tests.
Legacy descriptive phase D Review/collections	Scheduler, review sessions, collections, reset semantics, summaries.	Deterministic isolated queues.
Legacy descriptive phase E Audio/speech/media	Object storage/CDN, voices, generation/transcode, speech correct/incorrect/unavailable.	en-GB/en-US/cs-CZ + provider outage.
Legacy descriptive phase F Content/Admin	CMS, approvals, publication/rollback, content/media migrations, AI admin settings.	Draft-to-publish + rollback + AI permission tests.
Legacy descriptive phase G AI capability	Provider adapters, routing/fallback, learner AI, AI Assist/Auto allowlist, budgets/audit.	AI-off core passes; zero-paid-budget pass; provider fallback pass.
Legacy descriptive phase H Curriculum rollout	Module-by-module English/Czech A1-C1 through validated pipeline.	No placeholder levels; QA per module.
Legacy descriptive phase I Hardening/launch	Security/privacy/age, performance, SLO/alerts, backups, runbooks.	All acceptance scenarios pass.

39. Technical acceptance scenarios

ID	Scenario	Evidence
TAC-001	Bilingual start	New account completes representative English and Czech flows on same engine.
TAC-002	Language isolation	Mixed attempts/review/collections/AI memory preserve exact independent states.
TAC-003	Content-driven engine	Fixture swap changes content without renderer code edit; unknown schema recovers.
TAC-004	Media publication	Missing/unapproved asset blocks publish; approved
ID	**Scenario**	**Evidence**
		lesson works with generation provider offline.
TAC-005	Attempt durability	Double tap, retries, crash and analytics outage yield one authoritative effect.
TAC-006	Two-level mastery	Same attempt maps correctly to global skill and knowledge facets; reveal/hint differ.
TAC-007	Pronunciation UX	Usable speech -> correct/try again; low confidence/denial/provider failure -> speech unavailable with no penalty. _
TAC-008	Review	Incorrect/reveal/hint schedule differently; no cross-language candidate.
TAC-009	AI disabled	All core learner flows pass with every AI provider/feature disabled.
TAC-010	AI zero budget	Paid provider cannot be called while paid budget is zero/off.
TAC-011	AI fallback	Primary AI provider outage routes to approved fallback/local or deterministic non-AI fallback.
TAC-012	AI admin safety	AI Auto cannot publish/rollback/delete/change role/billing; allowed low-risk action is audited.

TAC-013	AI evidence safety	AI prose alone cannot update mastery; approved structured evaluator path can.
TAC-014	Responsive/a11y	320px, desktop, 200% zoom, keyboard, screen reader, reduced motion, AI streaming.
TAC-015	Publish/rollback	Active pinned run survives new publication and rollback without evidence loss.
TAC-016	Security/privacy	IDOR/RBAC/prompt-injection/log-redaction pass; voice/AI data lifecycle correct.
TAC-017	Originality/provenance	Published content/media and AI-generated assets have provenance and originality gate.
TAC-018	Operations/DR	Faults create correlated alerts; restore reproduces supported content/progress within RPO/RTO.
TAC-019	Evidence ownership isolation	Evidence referencing another user's attempt is rejected by composite FK/service guard; no mastery/review update occurs.
TAC-020	Evidence language isolation	Evidence/knowledge from a language different from attempt/profile is rejected; transaction leaves both language states unchanged.

40. Fixed decisions and ADR backlog

40.1. Fixed decisions carried forward into v1.7

- Lingvessa is the working product/repository name; brand legal clearance is separate before public launch.
- English and Čeština are real start target languages; uk-UA is UI/explanation locale; en-GB default and en-US selectable.
- Responsive Web/PWA first; modular monolith + workers; PostgreSQL source of truth; object storage/CDN for media.
- Two-level mastery: six global skill projections + knowledge-item facets.
- Usable speech uses learner-facing correct/try again; speech_unavailable is neutral and non-blocking.
- AI is optional; core course works without it. Manual/AI Assist/AI Auto; AI Auto cannot perform sensitive actions in v1.
- Paid AI disabled by default; initial paid AI budget = 0.
- Required lesson media is system-produced before publish; no user sourcing; no routine live-media dependency.
- Development default source/deploy: GitHub Private; Northflank Sandbox + Cloudflare R2 may be used for zero-cost reference implementation; production is provider-neutral.

40.2. ADRs required before named stage

ADR	Decision	Recommended default	Deadline
ADR-001	Production hosting/region/topology	Managed EU-capable services; choose after geography/privacy/budget.	Before production infra
ADR-002	Exact Next/Nest/Prisma majors	Current stable compatible with Node 24 LTS; pin after spike.	Repository baseline
ADR-003	Authentication implementation	Secure server sessions + Argon2id or reviewed managed auth.	Before identity
ADR-004	PostgreSQL HA/PITR tier	PG18 current minor; first-stage RPO/RTO as specified.	Before production data
ADR-005	Redis/queue	Postgres/outbox first; Redis-compatible adapter when queue/caching need measured.	Before scale jobs
ADR-006	Object storage/CDN	S3-compatible immutable; Cloudflare R2 dev default.	Before media stage
ADR-007	Image/TTS/STT providers	At least one adapter each; no provider shape in domain.	Before media integration
ADR-008	AI providers/models	Free/local-first; explicit approved list; paid off until budget decision.	Before AI capability rollout
ADR-009	AI privacy/retention/training settings	Per provider/task; data minimization.	Before real-user AI
ADR-010	Analytics/observability backend	OpenTelemetry-compatible; privacy allowlist.	Before production monitoring

ADR-011	Launch countries/minors/retention	No public registration until legal/product approval.	Before launch
ADR-012	Payments/subscription	Separate commercial/technical spec.	Before paid launch

41. Technical Definition of Done

- Code, DB migration, OpenAPI/content/event/AI policy schema and generated clients/types are versioned and reviewed.
- Unit/integration/contract/E2E plus relevant visual, accessibility, security, performance and fault tests pass.
- English and Czech fixtures traverse the same code path; language isolation includes AI context/memory.
- Two-level mastery mappings are explicit and regression-tested.
- Pronunciation states correct/incorrect/unavailable and Auto-advance behavior are regression-tested.
- AI-off core flow and zero-paid-budget tests pass.
- AI Assist/Auto actions have permissions, audit, budget/quota and safe fallback.
- No secrets, raw mic bytes or unapproved personal payload appear in logs/analytics/client bundle.
- Content/media/AI-generated artifact has schema lint, provenance, approvals, manifest and preview evidence.
- Documentation/ADR/spec/migrations/tests are updated together; clean checkout deploys through CI.

ДОДАТКИ

Додаток A. Attempt API example

POST /api/v1/lesson-runs/{run_id}/attempts

{

"attempt_id":"0191f2c0-7a6d-7c1e-9b02-1f3a4d5e6f70", "client_sequence":7,

"language_profile_id":"0198f0a0-aaaa-7abc-8def-aaaaaaaaaaaa", "language_profile_stable_code":"profile-cs", "content_version_id":"0198f0a0-bbbb-7abc-8def-bbbbbbbbbbbb", "version_code":"42",

"item_id":"0198f0a0-cccc-7abc-8def-cccccccccccc", "item_stable_code":"C-A1-M01-L02-A06", "response": {"kind":"text","value":"jsem"},

"assistance":{"hint_level":0,"revealed":false}

}

201 Created

{

"status":"correct", "assisted":false,

"feedback":{"code":"CORRECT","message_key":"feedback.correct"},

"effects":{"evidence_recorded":true,"review":{"scheduled":true,"due_at":"..."}},

"versions":{"evaluator":"...","mastery":"m2","scheduler":"s1"},

"correlation_id":"0198f0a0-2000-7abc-8def-200000000001"

}

Speech attempt uses status speech_correct | speech_incorrect | speech_unavailable. AI enhancement calls are separate resources and never masquerade as attempt success.

Додаток B. AI permission and fallback matrix

Action	Manual	AI Assist	AI Auto v1	Fallback
Personalized explanation	On explicit request	Allowed	Allowed if learner enabled	Deterministic authored explanation
Conversation/role-play	On explicit request	Allowed	Session auto-flow allowed	Scripted dialogue / no-AI route
Writing feedback	On explicit request	Allowed	Allowed with rubric/policy	Static rubric/self-check
Generate extra practice	Explicit	Suggest draft	Allowed only if validated ephemeral item	Curated review item

Content draft	Explicit	Allowed	Allowed low-risk draft creation	Manual authoring
Media generation	Explicit	Allowed	Allowed by quota/job policy	Existing approved asset/manual retry
Fix typo/metadata	Manual	Suggest diff	May auto-fix if allowlisted/reversible	Manual
Publish / rollback	Human only	AI may advise	FORBIDDEN	Human admin
Delete account / role / billing	Human only	AI may explain	FORBIDDEN	Human admin
Paid provider use	Only when enabled	Only when enabled	Only within explicit budget	Free/local or no-AI

Додаток C. Development deployment baseline (non-production)

Layer	Default	Status / reason
Source code	GitHub Private	All code/schemas/docs; unlimited private repos available on GitHub Free.
Web	Northflank Sandbox service 1	Always-on free sandbox; development/testing only.
API + dev worker	Northflank Sandbox service 2	May co-locate worker process for dev resource limit; production separation remains logical.
PostgreSQL	Northflank Sandbox addon	One free addon; sufficient for reference implementation, not production SLA.
Media	Cloudflare R2 Standard	S3-compatible; free tier includes 10 GB-month and free Internet egress as of 04.09.2026.
Redis	Not required initially	Adapter reserved; introduce managed Redis when measured queue/cache need justifies it.
AI	Free/local providers only by default	Paid budget 0; provider adapter/fallback required.

Portability rule

No application-domain contract may depend on Northflank, Cloudflare R2 or a specific AI vendor. OCI containers + PostgreSQL + S3-compatible storage + adapters are the portability boundary.

Додаток D. Traceability and sources

Source	Use
Formal Specification v1.7	Product requirements and precedence.
Technical Design v1.7	Detailed implementation contract for two-level mastery, speech state machine, relational schema and publication model.
AI Integration Specification v1.4	AI learning/admin capabilities, modes, provider abstraction, cost/privacy/fallback.
Curriculum Blueprint A1 v1.6	A1 pedagogy/module architecture; legacy pronunciation/mastery wording superseded where conflicting.
Reference Module Design v1.6	Executable English/Czech reference fixtures and 32-scenario regression baseline.
Node.js releases	Node.js releases - https://nodejs.org/en/about/previous-releases v24 LTS verified 04.09.2026.
PostgreSQL versioning	PostgreSQL versioning - https://www.postgresql.org/support/versioning/ PostgreSQL 18.6 current minor verified 04.09.2026.
Northflank pricing/docs	Northflank pricing/docs - https://northflank.com/pricing Developer Sandbox: 2 services, 1 addon, 2 jobs; development only.
Cloudflare R2 pricing	Cloudflare R2 pricing - https://developers.cloudflare.com/r2/pricing/ Standard free tier and free Internet egress.
GitHub pricing	GitHub pricing - https://github.com/pricing unlimited private repositories on Free.
W3C WCAG 2.2	W3C WCAG 2.2 - https://www.w3.org/TR/WCAG22/
OWASP ASVS	OWASP ASVS - https://owasp.org/www-project-application-security-verification-standard/
CEFR Companion Volume	Council of Europe (2020).

Handoff condition

This Technical Specification v1.7 supersedes Technical Specification v1.6 as the current technical baseline.

Technical Design v1.7 is the current implementation-detail authority; no future-version placeholder has precedence.

Curriculum/Reference docs saying speech is advisory-only or mastery is six-dimensions-per-item are superseded by the two-level mastery + binary speech safe-fallback contract in this document and Technical Design v1.7.

Part II. v1.7 consolidated controlled corrections (normative)

Controlled baseline synchronization - 09.09.2026

Normative baseline
This document is part of the synchronized Lingvessa owner-locked implementation baseline; P0 is authorized as of 10.09.2026. Earlier current-version references are superseded by the exact set below; historical "replaces" statements remain historical only.

Document	Version	Authority
Lingvessa_Formal_Specification	v1.7	Product scope, pedagogy, user-facing invariants
Lingvessa_Technical_Specification	v1.7	Normative technical requirements and interfaces
Lingvessa_Technical_Design	v1.7	Concrete implementation contracts, DDL/state/API design
Lingvessa_Curriculum_Blueprint_A1	v1.6	A1 pedagogical scope and module architecture
Lingvessa_Reference_Module_Design	v1.6	Executable E-A1-M01/C-A1-M01 fixtures
Lingvessa_AI_Integration_Specification	v1.4	AI-specific policy, data, routing, permissions
Lingvessa_Implementation_Plan	v1.4	Sole execution-order authority; no scope override

Decision	Topic	Fixed rule
BL-001	Baseline set	Formal 1.7 / Technical 1.7 / Design 1.7 / Curriculum A1 1.6 / Reference 1.6 / AI 1.4 / Plan 1.4
BL-002	Precedence	Domain-specific governing document controls; unresolved conflict -> owner decision + ADR/spec revision before code merge.
BL-003	Identity contract	id = UUID; stable_code = immutable human-readable semantic code; version_code = immutable revision label. Never put stable_code in *_id.
BL-004	Mastery contract	Exactly six global skills; canonical facet registry in this release. Reading is assessment-only domain, not seventh mastery skill.
BL-005	Language isolation	Composite DB FKs/checks/trigger guards enforce user + language_profile + language lineage for learner state and learner AI.
BL-006	Pronunciation states	speech_correct / speech_incorrect / speech_unavailable are canonical. Incorrect is non-blocking; Auto-advance occurs only after feedback dwell and never records success.
BL-007	Speech lifecycle	Asynchronous speech_job lifecycle with idempotent final attempt settlement and transient audio cleanup.
BL-008	AI API/data model	Canonical /api/v1/ai/explain|hint|practice|writing-feedback|conversations plus unified ai_* physical model.
BL-009	Monorepo	One /packages/provider-adapters package with ai/, speech/, tts/, image/, email/, storage/ sub-adapters; /api-

		client and /ui retained.
BL-010	Execution order	Implementation Plan v1.4 P0-P13 is the sole milestone numbering authority.
BL-011	Learner AI opt-in	Learner AI is OFF by default until explicit opt-in/consent. Effective enablement order is global -> capability -> provider/policy -> learner setting/consent -> quota.
BL-012	OpenAPI	Pin API description to OpenAPI Specification 3.1.2 for v1 baseline; upgrades require ADR/contract diff.

Precedence: Formal controls product/pedagogical scope; Curriculum controls A1 pedagogy; Technical Specification controls normative technical requirements; Technical Design controls concrete implementation detail; AI Specification controls AI-specific behavior; Reference Module Design is the executable fixture; Implementation Plan controls execution order only. Any unresolved conflict blocks implementation until an owner decision and controlled revision/ADR.

Canonical mastery/facet registry

Machine-code rule
Only facet codes in this registry may appear in mastery_facet/evidence_map. Every (skill, facet) target MUST also be listed in the eligible-pair matrix; pair mismatch blocks authoring/build/publish validation.

facet code	Meaning
recognition	Recognition of a known form/choice
meaning	Meaning comprehension
recall_l2_to_l1	Target-language -> explanation-language recall
recall_l1_to_l2	Explanation-language -> target-language recall
production	Independent target-language production
spelling	Spelling of lexical/form surface
orthography	Language-specific orthographic correctness
listening_recognition	Recognition from audio
listening_gist	Gist from audio
listening_detail	Specific detail from audio
pronunciation_target	Named pronunciation feature
intelligibility	Understandability of learner speech
context_use	Appropriate use in context
collocation	Chunk/collocation integrity
grammar_form	Grammar form
word_order	Word order
morphology	Inflection/morphology
register	Register/formality
variant_awareness	en-GB/en-US or accepted language variant awareness
interaction_repair	Repair/clarification move
interaction_goal	Goal completion in interaction

Facet-to-global-skill aggregation

Global skill	Eligible facets (only when evidence_map explicitly targets the skill)
Vocabulary	recognition, meaning, recall_l2_to_l1, recall_l1_to_l2, production, spelling, context_use, collocation, variant_awareness, register
Grammar	grammar_form, morphology, word_order, register, context_use
Listening	listening_recognition, listening_gist, listening_detail, meaning
Speaking	production, context_use, register, interaction_repair, interaction_goal,

	intelligibility
Writing	production, spelling, orthography, grammar_form, context_use, register
Pronunciation	pronunciation_target, intelligibility

Reading is not a seventh global mastery skill. Reading tasks may emit item-facet evidence to Vocabulary and/or Grammar according to the objective. The optional Level Test may report a separate assessment_domain=reading subscore; that subscore is assessment-only and is not persisted as skill_mastery_states.

Canonical identity and version contract

Field	Type	Example	Rule
id	UUID	018f...	Opaque database/API identity.
stable_code	string	E-A1-M01-L01	Immutable human-readable semantic code; never stored in *_id.
version_code	string	1.4 / 2026-09-05.1	Immutable revision label for a versioned entity.
provider_external_id	string nullable	vendor-model-x	External/provider identifier only in adapter/config boundary.

API/JSON examples MUST use UUID-shaped values in *_id fields. Content manifests MAY use stable_code for authoring references, but the build/publish step resolves them to UUID + version_code and rejects unresolved codes.

Database-enforced English/Czech isolation

Isolation invariant

Service-layer checks are not sufficient. Learner-state tables denormalize language_id where needed so PostgreSQL composite foreign keys can prove lineage.

language_profiles: UNIQUE(id,user_id,language_id), UNIQUE(user_id,language_id).
lessons/publications/content_versions/activity_items/knowledge_items: UNIQUE(id,language_id).
lesson_runs: explicit user_id + language_profile_id + language_id and composite FKs to language_profiles/publication/lesson.
activity_attempts: explicit user_id + language_profile_id + language_id; composite FK to lesson_runs; item/content FKs carry language_id.
evidence_events: explicit user_id + language_profile_id + language_id; composite FK to activity_attempts proves owner/language inheritance; knowledge/content FKs carry language_id; facet requires knowledge_item_id.
review/collections: composite profile/knowledge lineage. ai_runs: learner actor requires actor_id, language_profile_id and language_id plus composite FK to language_profiles.
DEFERRABLE constraint triggers cover remaining publication/item lineage. Negative fixtures for foreign-user and cross-language evidence are release-blocking.

Canonical pronunciation state and asynchronous speech-job contract

State	Meaning	Completion / evidence / Auto-advance
speech_correct	Usable, sufficiently confident target-feature match	Positive bounded Pronunciation evidence to the exact facet declared by evidence_map (pronunciation_target or intelligibility). Auto-advance allowed after feedback dwell if ON; otherwise show “Далі”.
speech_incorrect	Usable, sufficiently confident mismatch	Bounded negative/practice-needed evidence only to the exact Pronunciation facet declared by evidence_map (pronunciation_target or intelligibility). Non-blocking; Auto-advance may proceed after feedback dwell if ON; otherwise Retry + Continue; never success.
speech_unavailable	Mic denied/absent, timeout, low confidence, provider failure	No mastery update. Non-blocking. Auto-advance may proceed after neutral status dwell if setting ON; otherwise Retry + Continue.

Canonical state names are speech_correct, speech_incorrect and speech_unavailable. Legacy labels speech_feedback_correct and speech_feedback_retry are invalid.

speech_job lifecycle

created -> uploaded -> queued -> processing -> succeeded | unavailable | failed | cancelled | expired. POST /api/v1/speech/jobs with Idempotency-Key + attempt_id creates or returns the same owner/profile/item/content-bound job and returns job_id + single-use upload_url. PUT upload_url uploads encrypted transient audio; POST /api/v1/speech/jobs/{job_id}/submit transitions uploaded -> queued. GET /api/v1/speech/jobs/{job_id} returns owner-scoped status/result. POST /api/v1/speech/jobs/{job_id}/cancel is idempotent and settles the attempt exactly once as speech_unavailable(reason=user_cancelled); Continue is the normal LessonPlayer Next action after a terminal speech outcome and creates no new attempt. succeeded settles exactly once as speech_correct/speech_incorrect using the activity evidence_map facet; unavailable/failed/expired settle exactly once as speech_unavailable after bounded retry policy. Network retries reuse the same attempt_id/idempotency key/job; internal provider retries create no learner attempt; learner Retry after any terminal learner-facing result creates a NEW attempt_id and speech_job. Signed internal callbacks are owner/job/version validated; late/duplicate callbacks are logged and ignored after settlement. Temporary raw audio is deleted at terminal settlement/cancel or TTL <=15 minutes.

Canonical AI API and persistence mapping

API	Purpose
POST /api/v1/ai/explain	Grounded explanation
POST /api/v1/ai/hint	Bounded hint
POST /api/v1/ai/practice	Extra practice
POST /api/v1/ai/writing-feedback	Writing feedback
POST /api/v1/ai/conversations	Create bounded conversation session
POST /api/v1/ai/conversations/{session_id}/messages	Conversation turn
DELETE /api/v1/ai/conversations/{session_id}	Delete/close learner AI session
Logical concept	**Physical entity/table**
AIProviderProfile	ai_provider_profiles
AIModelProfile	ai_model_profiles
AIPolicy / permissions / budget eligibility	ai_policies
AIPromptTemplate	ai_prompt_templates
AIRequestRecord / AIExecution / AIRun	ai_runs
AISession	ai_sessions
AIMessage	ai_messages
AIArtifact	ai_artifacts
AIUsage / budget consumption	ai_usage_buckets + audit_events
AIToolAction	ai_tool_actions

Learner AI is OFF by default. Effective enablement order: global AI flag -> capability flag -> approved provider/model/data policy -> learner opt-in and required consent -> quota/budget. Any failed check returns deterministic fallback; no paid fallback is implicit.

Canonical monorepo package map

/apps/web /apps/admin /apps/api /apps/worker

/packages/domain /packages/application /packages/activity-schemas /packages/lesson-engine /packages/mastery-review /packages/ai-core /packages/provider-adapters # subfolders: ai/, speech/, tts/, image/, email/, storage/ /packages/api-client /packages/ui /packages/content-tools

/packages/provider-adapters/{ai,speech,tts,image,email,storage}/...

/content/english /content/czech /assets/specs /prisma /infra /docs/adr

There is no separate /packages/ai-adapters in the canonical baseline; AI adapters are a sub-area of /packages/provider-adapters. Provider SDK types remain inside adapter subpackages.

Canonical execution authority and milestone map

Milestone	Authority
P0	Governance freeze, exact baseline, decision log, full MUST trace matrix
P1	Monorepo/bootstrap/CI/config
P2	Data, UUID/stable_code/version_code, language isolation, publication
P3	Lesson Engine + attempts + deterministic state
P4	Two-level Mastery + Review + mistakes + collections
P5	Media/audio/speech + async speech jobs
P6	English E-A1-M01
P7	Czech C-A1-M01 + bilingual isolation proof
P8	Admin/CMS + publication + shared AI foundation + Admin AI Assist
P9	Optional learner AI + AI Auto allowlist (reuses P8 foundation)
P10	Onboarding/Home/Explore/Course/Progress/Profile/Settings/notifications + persistence
P11	Reference hardening + 32 scenarios + technical gates; freeze reference baseline
P12	A1 M02-M12 controlled rollout + pilot/calibration + optional A1 Level Test
P13	Full-system hardening/launch gate: security, privacy, performance, restore/DR, release provenance, legal/product readiness

Only Lingvessa_Implementation_Plan v1.4 uses P0-P13 numbering. Any older P-number labels in other specifications are descriptive history and MUST be mapped to these milestones rather than treated as execution authority.

OpenAPI baseline

The v1 implementation contract is pinned to OpenAPI Specification 3.1.2. References to “3.1.x” or /oas/latest.html are superseded. A later move to 3.2 requires an ADR, contract diff and generated-client compatibility gate.

Canonical external references

- OpenAPI Specification 3.1.2
- CEFR Companion Volume
- Council of Europe - CEFR
- English Profile
- Čeština pro cizince

OpenAPI here means the OpenAPI Specification for describing HTTP APIs; it is unrelated to OpenAI as an AI provider. OpenAI remains only a possible future provider behind the provider-agnostic AI adapter and is not required by the v1 baseline.

Approval record

Поле	Значення
Baseline status	OWNER-LOCKED IMPLEMENTATION BASELINE / P0 AUTHORIZED
Approval date	10.09.2026
Approver	Product owner
Candidate version	v1.7

Accepted exceptions	None
Execution authority	Lingvessa_Implementation_Plan v1.4; P0 must be closed before P1.

Canonical machine-readable source: /packages/activity-schemas/mastery-registry.json. CI MUST validate every evidence_map against the registry and eligible skill-facet pair matrix; copied tables in specifications are checksum-checked mirrors.